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Surname:															
First Name(s):															

Department of Mathematics and Applied Mathematics
MAM2000W: 2LA (Linear Algebra)

June exam	1 June 2022
Time : 2 hours	Maximum marks: 62
	Full marks: 60

- This question paper consists of 15 pages (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. We won't take this work in! Only the answers that you write on this question paper will be marked.
- Basic scientific calculators also allowed.
- For Section A, only final answers will be marked. For Section B, full answers must be given.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

DO NOT WRITE BELOW THIS LINE

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	TOTAL
7	10	6	4	10	12	8	5	62

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Section A

In this section, give your final answer only. Do not show your working.

Question 1. [7 points] Let

$$A = \begin{pmatrix} 2 & 0 & 1 & 3 & -1 \\ -4 & -1 & 0 & -6 & 2 \\ 0 & 3 & -6 & 3 & 1 \\ 2 & -1 & 3 & 6 & 0 \end{pmatrix}, \quad U = \begin{pmatrix} 2 & 0 & 1 & 3 & -1 \\ 0 & -1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The matrix U is a row-echelon form of the matrix A (you need not check this).

(a) Give the rank and nullity of A .

Rank Nullity

(b) Give a basis for

(i) $NS(A)$

(ii) $CS(A)$

(c) Which one of the following statements is true?

- (i) The system $A\mathbf{x} = \mathbf{b}$ has a unique solution for all $\mathbf{b} \in \mathbb{R}^4$.
- (ii) The system $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions for all $\mathbf{b} \in \mathbb{R}^4$.
- (iii) The system $A\mathbf{x} = \mathbf{b}$ has no solution for some $\mathbf{b} \in \mathbb{R}^4$.
- (iv) The system $A\mathbf{x} = \mathbf{b}$ has a unique solution for some $\mathbf{b} \in \mathbb{R}^4$.
- (v) The system $A\mathbf{x} = \mathbf{b}$ has a no solution for all $\mathbf{b} \in \mathbb{R}^4$.

Give the number of the correct statement:

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Question 2. [10 points] Let $A = \begin{pmatrix} 4 & -3 & -3 \\ 2 & -3 & -1 \\ 0 & 0 & -2 \end{pmatrix}$.

(a) Find all the eigenvalues of A .

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(b) Is A invertible? Yes

☐

No

☐

Give a reason for your answer using only your answer to (a).

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(c) For each eigenvalue of A , find a basis for the corresponding eigenspace.

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(d) Is A diagonalizable? Yes

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No

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Give a reason for your answer.

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(e) If A is diagonalizable, find an invertible matrix P and a diagonal matrix D such that $P^{-1}AP = D$. If it is not, find a basis B for $\mathbb{R}^3$ containing at least two eigenvectors of A .

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Question 3. [6 points] For $p(x), q(x) \in P_2$, put $\langle p(x), q(x) \rangle = \int_0^1 p(x)q(x) dx$. You may assume that this defines an inner product on P_2 . Let $p(x) = x - 2$ and $q(x) = 2x^2$.

(a) Find $\langle p(x), q(x) \rangle$.

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(b) Find $\|q(x)\|$.

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(c) For which value of b will $r(x) = x - b$ be orthogonal to $q(x)$?

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Section B

In this section, give full answers to all the questions. Show all your working.

Question 4. [4 points] Let $A = \begin{pmatrix} 1 & 2 \\ -3 & 4 \end{pmatrix}$. Write A as a product of elementary matrices.

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Question 5. [10 points] The subset S of $M_{2 \times 2}(\mathbb{R})$ is given by

$$S = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 2 & 2 \\ 1 & 0 \end{pmatrix} \right\}.$$

Answer the following questions, in each case showing clearly how you arrived at your answer.

(a) Is S linearly independent?

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(b) Is $\text{span}(S) = M_{2 \times 2}(\mathbb{R})$?

If it is, explain why it is, or if it is not, find an equation in a, b, c and d that all matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ in $\text{span}(S)$ must satisfy.

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(c) Write down a basis B for $\text{span}(S)$.

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(d) Is B a basis for $M_{2 \times 2}(\mathbb{R})$? Give a reason for your answer. If not, find a matrix A such that $B \cup \{A\}$ is a basis for $M_{2 \times 2}(\mathbb{R})$ and explain how you chose the matrix.

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- (c) Find the matrix ${}_C[T]_B$ representing T with respect to the basis B in P_2 and the standard basis C for $M_{2 \times 2}(\mathbb{R})$.

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- (d) Find the kernel of T .

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(e) Find a basis for the range of T .

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(f) is T a linear isomorphism? Give a reason for your answer.

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Question 7. [8 points] Read the following result and its proof and then answer the questions following it.

Theorem Let V be an n -dimensional vector space and $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a linearly independent subset of V . Then S is a basis for V .

Proof Suppose S is not a basis for V (1)

Then there is a $\mathbf{v} \in V$ such that $\mathbf{v} \notin \text{span } S$ (2)

Suppose $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n + \lambda_{n+1} \mathbf{v} = \mathbf{0}$ (3)

If $\lambda_{n+1} \neq 0$, then $\mathbf{v} = \frac{-1}{\lambda_{n+1}}(\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n)$, (4)

a contradiction. Hence we must have $\lambda_{n+1} = 0$ (5)

If we substitute this in (3), it follows from the resulting equation that

$\lambda_1 = \lambda_2 = \dots = \lambda_n = 0$ (6)

It follows that $S \cup \{\mathbf{v}\}$ is linearly independent. (7)

But this contradicts the fact that V is n -dimensional. (8)

Therefore S must be a basis for V .

(a) What method of proof is used here?

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(b) Explain how line (2) follows from line (1).

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(c) What is the contradiction in line (4)?

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(d) Explain why we can conclude in line (6) that $\lambda_1 = \lambda_2 = \cdots = \lambda_n = 0$.

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(e) Why can we say in line (7) that $S \cup \{\mathbf{v}\}$ is linearly independent?

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(f) Why does this contradict the fact that V is n -dimensional? (You may quote a theorem here.)

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- (g) Let T be a subset of an n -dimensional vector space V such that $\text{span}(T) = V$, but T is not a basis of V . What can be said about the number of elements in T ? Give a reason for your answer.

Question 8. [5 points] We will call a square matrix A *balanced* if $NS(A) = CS(A)$. Prove or disprove:

- (a) There is a 2×2 balanced matrix.

